# A CRITERION FOR BIRATIONALITY OF INTEGRAL SCHEMES VIA RESIDUE FIELDS OF GENERIC POINTS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Editorial note.* Read in full under the Lead's 12:4x rule. The proof contains 'It is immediate that (2) implies (3)' - a genuinely immediate implication between two of the listed conditions; the rest of the argument is given in full, so this is not a hint-only proof.

*Allowed standard prerequisites (editorial).*

- Equivalence of birationality with the existence of isomorphic nonempty opens. Lemma \ref{lemma-birational-integral}, cited at proof lines 13160-13161. Gives the first equivalence.
- The local ring at a generic point of a reduced scheme is its residue field. Algebra, Lemma \ref{algebra-lemma-minimal-prime-reduced-ring}, cited at proof lines 13164-13166. Turns the residue field condition into the local ring condition.
- Existence of a rational map realising a prescribed isomorphism, and the birational-morphism criterion. Lemma \ref{lemma-rational-map-finite-presentation} and Lemma \ref{lemma-birational-birational}, cited at proof lines 13167-13173. Produce the morphism and conclude.

*Source reference.* Original lemma 0552: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6660–6660. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 0BAA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14203–14203. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 00EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 548–548. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0BX8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14971–14971. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 0BAC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14205–14205. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $S$ be a scheme. Let $X$ and $Y$ be integral schemes locally of finite type over $S$. Let $x \in X$ and $y \in Y$ be the generic points. The following are equivalent*

1. *$X$ and $Y$ are $S$-birational,*
2. *there exist nonempty opens of $X$ and $Y$ which are $S$-isomorphic, and*
3. *$x$ and $y$ map to the same point $s \in S$ and $\kappa(x) \cong \kappa(y)$ as $\kappa(s)$-extensions.*

*Proof.* We have seen the equivalence of (1) and (2) in Lemma 0BAA. It is immediate that (2) implies (3). To finish we assume (3) holds and we prove (1). Observe that $\mathcal{O}_{X,x} = \kappa(x)$ and $\mathcal{O}_{Y,y} = \kappa(y)$ by Algebra, Lemma 00EU. By Lemma 0BX8 there is a rational map $f : U \to Y$ which sends $x \in U$ to $y$ and induces the given isomorphism $\mathcal{O}_{Y,y} \cong \mathcal{O}_{X,x}$. Thus $f$ is a birational morphism and hence induces an isomorphism on nonempty opens by Lemma 0BAC. This finishes the proof. □